

Lecture 3.1

SLLN

Intro to Expectation Limit Theorems

Last Time

- Prove $(X_n \xrightarrow{a.s.} Y) \Rightarrow (X_n \xrightarrow{P} Y)$
- Examples
- $(Z_n \xrightarrow{a.s.} z) \iff (\forall \epsilon > 0, P(|Z_n - z| \geq \epsilon \text{ i.o.}) = 0)$
- $(\forall \epsilon > 0, \sum_{n=1}^{\infty} P(|Z_n - z| \geq \epsilon) < \infty) \iff (Z_n \xrightarrow{a.s.} z)$

Proof of SLLN

Strong Law of Large Numbers (SLLN)

If $X_1, X_2, X_3, \dots$ indep, each with finite mean m , and $E((X_i - m)^4) \leq a < \infty \forall i$,

then $\frac{1}{n}(X_1 + X_2 + \dots + X_n) \xrightarrow{a.s.} m$

Proof:

Let $Y_i = X_i - m$, then $E[Y_i] = E[X_i] - m = 0$
 $E[\bar{Y}_n] = E[Y_i] = 0$

Also, we can see that,

$$\begin{aligned}\bar{Y}_n &= \frac{1}{n}(Y_1 + \dots + Y_n) = \frac{1}{n}(X_1 - m + X_2 - m + \dots + X_n - m) \\ &= \frac{1}{n}(X_1 + X_2 + \dots + X_n) - m\end{aligned}$$

In this way, when we prove

$$\boxed{\bar{Y}_n \xrightarrow{a.s.} 0}$$

we have,

$$\frac{1}{n}(X_1 + X_2 + \dots + X_n) - m \xrightarrow{a.s.} 0$$

then,

$$\bar{X}_n \xrightarrow{a.s.} m$$

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Now, we want to show

$$\bar{Y}_n \xrightarrow{a.s.} 0 \quad \text{and} \quad Y_i = X_i - m$$

we know $E[(X_i - m)^4] \leq a < \infty$, and by observing

$$(X_i - m)^2 \leq (X_i - m)^4 + 1$$

In this way,

$$E[(X_i - m)^2] \leq E[(X_i - m)^4] + 1 \leq a + 1 \equiv v < \infty$$

Same as,

$$E[Y_i^2] \leq E[Y_i^4] + 1 \leq a + 1 \equiv v < \infty$$

Now, we define

$$S_n = Y_1 + Y_2 + \dots + Y_n, \quad \text{then} \quad \bar{Y}_n = \frac{S_n}{n}$$

We consider $E(S_n^4)$

$$E(S_n^4) = E[(Y_1 + \dots + Y_n)(Y_1 + \dots + Y_n)(Y_1 + \dots + Y_n)(Y_1 + \dots + Y_n)]$$

~~$E(S_n^4)$~~

S_n^4 contains many terms. $\textcircled{1} Y_i Y_j Y_k Y_l$ (i, j, k, l distinct)

$$\text{-----} \textcircled{2} Y_i Y_j Y_k^2$$

$$\text{-----} \textcircled{3} Y_i Y_j^3$$

The Expected Value for $\textcircled{1}, \textcircled{2}, \textcircled{3}$ are all zero because,

$$\textcircled{1} \quad E[Y_i Y_j Y_k Y_l] = E[Y_i] E[Y_j] E[Y_k] E[Y_l] = 0$$

$$\textcircled{2} \quad E[Y_i Y_j Y_k^2] = E[Y_i] E[Y_j] E[Y_k^2] = 0$$

The only terms left are

$$\text{and} \quad \binom{n}{2} \binom{4}{2} Y_i^2 Y_j^2$$

we remember $E[Y_i^4] \leq a$

$$E[Y_i^2] \leq v$$

$$\binom{n}{2} \binom{4}{2}$$

$$= \frac{n!}{2!(n-2)!} \frac{4!}{2!2!} = 3n(n-1)$$

In this way,

$$E[S_n^4] = nE[Y_1^4] + 3n(n-1)E[Y_1^2 Y_2^2]$$

$$\leq na + 3n(n-1)v^2$$

$$\leq n^2 a + 3n^2 v^2 = (3v^2 + a)n^2 = kn^2$$

$$\text{when } k = 3v^2 + a$$

Finally,

$$\forall \epsilon > 0, P(|\bar{Y}_n - 0| \geq \epsilon)$$

$$= P\left(\left|\frac{S_n}{n}\right| \geq \epsilon\right) = P(|S_n| \geq n\epsilon)$$

$$= P(|S_n|^4 \geq n^4 \epsilon^4) \leq \frac{E[S_n^4]}{n^4 \epsilon^4}$$

$$\leq \frac{kn^2}{n^4 \epsilon^4} = \frac{k}{\epsilon^4} \cdot \frac{1}{n^2}$$

In this way,

$$\forall \epsilon > 0, \sum_{n=1}^{\infty} P(|\bar{Y}_n - 0| \geq \epsilon) \leq \sum_{n=1}^{\infty} \frac{k}{\epsilon^4} \cdot \frac{1}{n^2} < \infty$$

$$\therefore \bar{Y}_n \xrightarrow{\text{a.s.}} 0$$

$$\text{i.e. } \bar{X}_n - m \xrightarrow{\text{a.s.}} 0$$

$$\text{i.e. } \bar{X}_n \xrightarrow{\text{a.s.}} m$$

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Fact.

If $\{X_n\}$ i.i.d., they don't even need $a < \infty$.

(Hence also WLLN)

[Rig § 5.4.]

Example (Fair coin tossing)

$$X_n = \begin{cases} 1, & \text{nth coin Heads} \\ 0, & \text{nth coin Tails} \end{cases}$$

$$M_n = \frac{1}{n}(X_1 + X_2 + X_3 + \dots + X_n)$$

$$\text{WLLN: } M_n \xrightarrow{P} \frac{1}{2}, \text{ so, e.g. } P(M_n < 0.503) > 0.99$$

$\forall$ (large enough n)

$$\text{SLLN: } M_n \xrightarrow{\text{a.s.}} \frac{1}{2}, \text{ so, e.g. } P(M_n < 0.503 \mid \text{large enough } n) = 1$$

Intro Expectation Limit Theorems

Motivation: suppose $X_n \xrightarrow{a.s.} X$, Does this imply that $E[X_n] \rightarrow E[X]$?

需要一些条件
e.g. $P(s) = 2^{-s}$, $s \in \mathbb{N}$ sample space 为 $\mathbb{N}$

$$\text{suppose } X_n(s) = 2^n \int_{n,s} = \begin{cases} 2^n & n=s \\ 0 & \text{o.w.} \end{cases}$$

Then, $\lim_{n \rightarrow \infty} X_n(s) = 0$, since $X_n(s) = 0 \quad \forall n > s$

e.g. $s=4$

$$X_1(4)=0, X_2(4)=0, X_3(4)=0, X_4(4)=16, X_5(4)=0, X_6(4)=0, \dots$$

So we have

$$X_n \xrightarrow{a.s.} 0$$

However,

$$E[X_n] = \sum_{s \in \mathbb{N}} P(s) X_n(s) = 0 + \sum X_n(n) P(n)$$

$$= 2^n \cdot 2^{-n} = 1$$

$$\therefore E[X_n] \not\rightarrow E[0]$$

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Lecture 3.2

Expectation Limit Theorem

Bounded Convergence Thm

Monotone Convergence Thm

Motivation: 继续讨论 when $E[X_n] \rightarrow E[X]$?

Bounded Convergence Theorem

If $X_n \xrightarrow{a.s.} X$, and $\exists k < \infty$ with $|X_n| \leq k \quad \forall n$
i.e. $|X_n(s)| \leq k \quad \forall n, \forall s \in S$,

then, $\lim_{n \rightarrow \infty} E[X_n] = E[X]$

Proof:

$$|E(X) - E(X_n)| = |E(X - X_n)| \leq E(|X - X_n|)$$

"generalized triangle inequality"
 $|E(Y)| \leq E(|Y|)$

Let $\varepsilon > 0$, and $A_n = \{|X - X_n| > \varepsilon\}$

Then,

$$|X - X_n| \leq \begin{cases} \varepsilon & \text{on } A_n^c \\ 2k & \text{on } A_n \end{cases} \leq \varepsilon + 2k \mathbb{1}_{A_n}$$

$$|X - X_n| \leq 2k$$

$$\text{Since } |X - X_n| \leq |X| + |X_n|$$

$$\leq 2k$$

$$(|X| \leq k \text{ because } |X_n| \leq k \text{ and } X_n \xrightarrow{a.s.} X)$$

same thing

Then, $\limsup_{n \rightarrow \infty} E(|X - X_n|)$

$$\leq \limsup_{n \rightarrow \infty} E(\varepsilon + 2k \mathbb{1}_{A_n}) = \varepsilon + 2k \limsup_{n \rightarrow \infty} E[\mathbb{1}_{A_n}]$$

$$\mathbb{1}_{A_n} = \begin{cases} 1 & \text{when } A_n \\ 0 & \text{when } A_n^c \end{cases}$$

Lemma
 $\limsup_{n \rightarrow \infty} P(A_n) \leq P(A_n \text{ i.o.})$
 from lecture one.

$$= \varepsilon + 2k \limsup_{n \rightarrow \infty} P(A_n)$$

$$\leq \varepsilon + 2k P(A_n \text{ i.o.})$$

$$= \varepsilon + 2k \cdot 0$$

$$= \varepsilon$$

$$P(A_n \text{ i.o.}) = 0$$

since $X_n \xrightarrow{a.s.} X$
 $\Leftrightarrow \forall \varepsilon > 0, P(|X_n - X| \geq \varepsilon \text{ i.o.}) = 0$
 $P(A_n \text{ i.o.}) = 0$

Then,

$$\limsup_{n \rightarrow \infty} E(|X - X_n|) \leq \varepsilon \quad \forall \varepsilon$$

$$\Rightarrow E(|X - X_n|) \rightarrow 0$$

$$\Rightarrow |E(X) - E(X_n)| \leq E(|X - X_n|) \rightarrow 0$$

$$\Rightarrow E(X_n) \rightarrow E(X)$$

□

Monotone Convergence Theorem

If $X_1, X_2, \dots$ are non-negative r.v., and $\{X_n\} \uparrow X$,

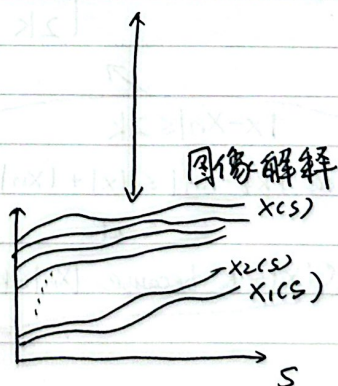
$$\text{i.e. } 0 \leq X_1 \leq X_2 \leq X_3 \leq \dots$$

$$\text{i.e. } 0 \leq X_1(s) \leq X_2(s) \leq \dots \quad \forall s \in S$$

and $X_n \xrightarrow{a.s.} X$.

Then,

$$\lim_{n \rightarrow \infty} E(X_n) = E(X)$$



$\forall s \in S, (\text{fixed } s)$

$$0 \leq X_1(s) \leq X_2(s) \leq \dots$$

Proof:

Here,

$$\{X_n\} \uparrow X \Rightarrow 0 \leq E(X_1) \leq E(X_2) \leq E(X_3) \leq \dots \leq E(X)$$

Then, (by the monotone sequence theorem)

$$\lim_{n \rightarrow \infty} E(X_n) \text{ exists and } \leq E(X) \leftrightarrow \text{though both could be } +\infty,$$

So we have,

$$\lim_{n \rightarrow \infty} E(X_n) \leq E(X)$$

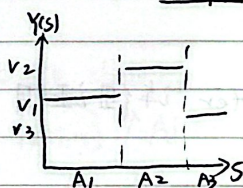
Now we only need to show

$$\lim_{n \rightarrow \infty} E(X_n) \geq E(X)$$

To prove this, we need new definition of $E(X)$.

defn.1

r.v. Y is simple if $Y = \sum_{i=1}^n v_i \mathbb{I}_{A_i}$ where A_i is a partition of S .

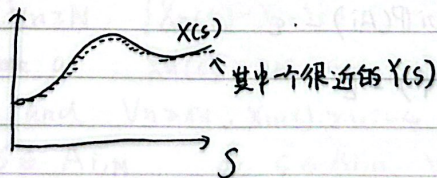


$$\text{we can see that } E(Y) = \sum_{i=1}^n v_i P(A_i)$$

Then we can have a new definition for $E(X)$, $X \geq 0$

defn.2

For any r.v. $X \geq 0$, $E(X) = \sup \{E(Y) : Y \text{ simple}; Y \leq X\}$



From last page, we have

① WTS $\lim_{n \rightarrow \infty} E(X_n) \geq E(X)$

② Y is simple $\Leftrightarrow Y = \sum_{i=1}^m v_i \mathbb{1}_{A_i}$ $\{A_i\}$ partition

③ $E(X) = \sup \{E(Y) : Y \text{ simple}, Y \leq X\}$ ← new def
 $X \geq 0$

In this way, it suffices to show.

$\lim_{n \rightarrow \infty} E(X_n) \geq E(Y) \quad \forall Y \text{ is simple } Y \leq X$
 i.e. $Y = \sum_{i=1}^m v_i \mathbb{1}_{A_i} \quad Y \leq X$

Let $\varepsilon > 0$, and

$A_{i,n} = \{s \in A_i : X_n(s) \geq v_i - \varepsilon\}$

Then, (since $\{X_n\} \uparrow X$)

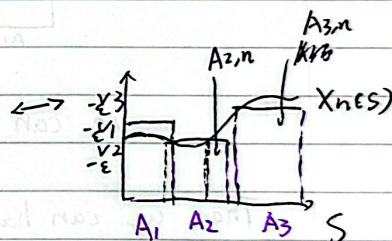
$A_{i,n} \subseteq A_{i,n+1}$

Also, ($\{X_n\} \uparrow X \geq v_i$ on A_i)

$\{A_{i,n}\} \uparrow A_i$

Later 详细证明

Then $E(X_n) \geq \sum_{i=1}^m (v_i - \varepsilon) P(A_{i,n})$ since $X_n \geq \sum_{i=1}^m v_i \mathbb{1}_{A_{i,n}}$



Then, (let $n \rightarrow \infty$)

$\lim_{n \rightarrow \infty} E(X_n) \geq \sum_{i=1}^m (v_i - \varepsilon) P(A_i)$

← $\lim_{n \rightarrow \infty} P(A_{i,n}) = P(\bigcup_{n=1}^{\infty} A_{i,n}) = P(A_i)$

$= \sum_{i=1}^m v_i P(A_i) - \varepsilon \sum_{i=1}^m P(A_i)$

$= \sum_{i=1}^m v_i P(A_i) - \varepsilon$

$= E(Y) - \varepsilon$

← $\sum_{i=1}^m P(A_i) = 1$ ← partition

Then,

$\lim_{n \rightarrow \infty} E(X_n) \geq E(Y) - \varepsilon \quad \forall \varepsilon$

$\therefore \lim_{n \rightarrow \infty} E(X_n) \geq E(Y) \quad \forall \text{ simple } Y, Y \leq X$

$\therefore \lim_{n \rightarrow \infty} E(X_n) \geq E(X) \Rightarrow \lim_{n \rightarrow \infty} E(X_n) = E(X)$

Later 详细证明部分

Let $\varepsilon > 0$, and

$$A_{i,n} = \{s \in A_i : X_n(s) \geq v_i - \varepsilon\} \quad (\{X_n\} \uparrow X)$$

① We want to show

$$A_{i,n} \subseteq A_{i,n+1}$$

Pf:

Let, $s \in A_{i,n}$, then we know

$$s \in A_i \text{ and } X_n(s) \geq v_i - \varepsilon$$

Since $\{X_n\} \uparrow X$, we have

$$X_{n+1}(s) \geq X_n(s) \geq v_i - \varepsilon$$

$$\Rightarrow s \in A_{i,n+1}$$

$$\Rightarrow A_{i,n} \subseteq A_{i,n+1} \quad \square$$

② We want to show

$$\{A_{i,n}\} \uparrow A_i$$

We already know $A_{i,n} \subseteq A_{i,n+1}$ i.e. $\{A_{i,n}\} \uparrow \bigcup_{n=1}^{\infty} A_{i,n}$

We only need

$$A_i = \bigcup_{n=1}^{\infty} A_{i,n}$$

$$\text{i.e. } A_i \subseteq \bigcup_{n=1}^{\infty} A_{i,n} \quad \text{since } A_{i,n} \subseteq A_i \quad \forall n$$

Let $s \in A_i$, (since $\gamma(s) = v_i$ and $X \geq \gamma$)

$$\therefore X(s) \geq v_i \quad \Rightarrow X(s) \geq v_i > v_i - \varepsilon$$

$$\therefore X(s) > v_i - \varepsilon$$

Also, since $\{X_n\} \uparrow X$, we can define $\varepsilon_2 = \frac{X(s) - (v_i - \varepsilon)}{2}$, and $\exists N$ great enough s.t.

$$\forall n \geq N, |X_n(s) - X(s)| \leq \varepsilon_2$$

$$\text{Same as } X_n(s) > v_i - \varepsilon$$

Since $s \in A_i$ and $\forall n \geq N, X_n(s) > v_i - \varepsilon$,

$$s \in A_{i,n} \quad \text{or } s \in A_{i,n} \quad \forall n \geq N$$

$$\therefore A_i \subseteq \bigcup_{n=1}^{\infty} A_{i,n}$$

$$\boxed{\therefore A_i = \bigcup_{n=1}^{\infty} A_{i,n}} \quad \square$$

Fatou's Lemma

If $X_n \geq c \quad \forall n$, where $c > -\infty$

Then

$$E(\liminf_{n \rightarrow \infty} X_n) \leq \liminf_{n \rightarrow \infty} E(X_n)$$

[By replacing X_n by $X_n - c$, suffices to assume $c = 0$]

↑

我们只需证明 $(X_n \geq 0 \quad \forall n)$

$$\Rightarrow (E(\liminf_{n \rightarrow \infty} X_n) \leq \liminf_{n \rightarrow \infty} E(X_n))$$

why?

If we already prove $(X_n \geq 0 \quad \forall n) \Rightarrow (\text{---})$,
then

$$\forall X_n \geq c \quad \forall n, (c > -\infty)$$

we can take

$$Y_n = X_n - c, \text{ then } Y_n \geq 0 \quad \forall n$$

Then by the theorem,

$$E(\liminf_{n \rightarrow \infty} Y_n) \leq \liminf_{n \rightarrow \infty} E(Y_n)$$

$$\Rightarrow E(\liminf_{n \rightarrow \infty} X_n) - c \leq \liminf_{n \rightarrow \infty} E(X_n) - c$$

$$\Rightarrow E(\liminf_{n \rightarrow \infty} X_n) \leq \liminf_{n \rightarrow \infty} E(X_n)$$

Recall 之前的例子, 在 Fatou's Lemma 的体现

e.g. $P(s) = 2^{-s} \quad \forall s \in \mathbb{N} \quad X_n(s) = 2^n \mathbb{I}_{n \leq s}$

对于 Fatou's Lemma 来说,

$$\therefore \text{LHS} \leq \text{RHS}$$

$$\lim_{n \rightarrow \infty} X_n = 0 \Rightarrow \text{LHS} = E(\liminf_{n \rightarrow \infty} X_n) = E(0) = 0$$

$$\lim_{n \rightarrow \infty} E(X_n) = \sum_{s \in \mathbb{N}} P(s) X_n(s) = 1 \Rightarrow \text{RHS} = \liminf_{n \rightarrow \infty} E(X_n) = \lim_{n \rightarrow \infty} (1) = 1$$

e.g. Instead $X_n(s) = -2^n \int_n s$

$$\text{LHS} = 0$$

$$\text{RHS} = -1$$

$$\Downarrow$$
$$\liminf_{n \rightarrow \infty} \int_s P(s) X_n(s) = -2^n \cdot 2^{-n} = -1$$

$$\text{LHS} \geq \text{RHS}$$

why?

Not bounded below.

We need $X_n \geq c \forall n$, where $c > -\infty$.

i.e. $\exists c > -\infty$ s.t. $X_n(s) \geq c \forall n, \forall s$

But $\boxed{-2^n \int_n s} \forall n, \forall s$ can't be bounded below